

Pharmacokinetics and Pharmacodynamics of Mepolizumab, an Anti-Interleukin-5 Monoclonal Antibody

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Abstract

Mepolizumab is a fully humanized monoclonal antibody (IgG1/ κ) targeting human interleukin-5 (IL-5), a key haematopoietin needed for eosinophil development and function. Mepolizumab blocks human IL-5 from binding to the α -chain of the IL-5 receptor complex on the eosinophil cell surface, thereby inhibiting IL-5 signalling. The pharmacokinetics of mepolizumab have been evaluated in clinical studies at doses of 0.05–10 mg/kg and at 250 mg, 750 mg and 1500 mg. Mepolizumab was eliminated slowly, with mean initial and terminal phase half-life values of approximately 2 and 20 days, respectively. Plasma clearance ranged from 0.064 to 0.163 mL/h/kg and steady-state volume of distribution ranged from 49 to 93 mL/kg. Pharmacokinetics were dose proportional and time independent. Estimates based on a two-compartment intravenous infusion model from patients with asthma or healthy subjects following single doses predicted mepolizumab plasma concentrations in multiple-dose studies involving patients with hypereosinophilic syndrome (HES), asthma or eosinophilic oesophagitis. The absolute bioavailability of mepolizumab was 64–75% following subcutaneous injection and 81% following intramuscular injection. Peripheral blood eosinophil levels decreased in healthy subjects and patients with HES, asthma, eosinophilic oesophagitis or atopic dermatitis after intravenous mepolizumab infusion and subcutaneous injection. Reductions in eosinophil counts in oesophagus, sputum, skin, bone marrow, nasal lavage fluid and/or bronchial mucosa after

treatment with mepolizumab were observed in placebo-controlled studies in various indications. The relationship between percentage change from baseline in blood eosinophils and mepolizumab plasma concentrations was described by an indirect pharmacological response model. The estimated maximal decrease in eosinophil count was approximately 85% from baseline and the half-maximal inhibitory concentration (IC_{50}) was approximately 0.45 $\mu\text{g/mL}$.

Eosinophil development is primarily controlled by interleukin (IL)-3, IL-5 and granulocyte-macrophage colony stimulating factor (GM-CSF). Of these, IL-5 is most selective for eosinophils and is the major cytokine responsible for regulating eosinophil growth, differentiation, mobilization, recruitment, activation and survival.^[1-4] The IL-5 receptor complex is expressed on the eosinophil cell surface. This receptor comprises a ligand-specific α -chain and a β -chain, which is the signal-transducing unit and is identical to the β -chain of the GM-CSF and IL-3 receptors.^[5] Basophils also express IL-5 receptors, and IL-5 enhances the release of mediators such as histamine and leukotriene C4 from these cells.^[6,7] However, IL-5 is not obligatory for this function and IL-3 is more potent than IL-5 at mediating the survival and activation of human basophils.^[8] IL-5 has limited effects on other cell types.^[1,9,10]

Various stimuli induce the recruitment of eosinophils from the circulation to sites of inflammation, where they modulate the immune response through pleiotropic mediators.^[9] Tissue damage and dysfunction resulting from secretion of these mediators is a common pathogenic component in several eosinophil-driven diseases, including asthma, atopic dermatitis, eosinophil-associated gastrointestinal disorders (eosinophilic oesophagitis, eosinophilic gastritis, eosinophilic gastroenteritis) and hypereosinophilic syndrome (HES).

1. Treatment Issues

Eosinophil-associated disease progression is dependent on increased eosinophil production and release from the bone marrow followed by selective recruitment into sites of disease activity. The objective of treatment is long-term disease control and symptom relief rather than repeated remission induction with aggressive therapies. However, current maintenance therapies for patients with eosinophilic disorders are far from ideal. Treatments alleviating symptoms do not tend to affect underlying disease processes and their efficacy is limited or may trigger the development of resistance. Immunosuppressive and immunomodulatory agents, such as topical/systemic corticosteroids, are associated with significant long-term side effects. Cytotoxic agents (e.g. hydroxyurea and vincristine), interferon- α and imatinib (which is first-line treatment for *FIP1L1/*

PDGFRA-positive HES) are used in HES patients, and all are poorly or less tolerated than mepolizumab.^[11-16] Because of their tolerability profiles, these drugs may have to be given at reduced doses or discontinued, increasing the risks of disease relapse and organ damage. Management of eosinophilic oesophagitis relies largely on systemic/topical corticosteroids and dietary therapies, although proton pump inhibitors, leukotriene receptor antagonists, immunomodulatory agents and endoscopic dilatation are also used.^[17-19]

IL-5 inhibition is a new and logical therapeutic target for eosinophilic disorders. Given the critical and selective role of this cytokine in regulating eosinophil development and function, anti-IL-5 therapies can be expected to attenuate eosinophil maturation and trafficking into target tissues, and to inhibit the associated adverse sequelae without affecting the normal activities of other cells of the immune system.^[20,21] Mepolizumab (GlaxoSmithKline) and reslizumab (Ception Therapeutics, Inc.) represent this therapeutic approach.^[11,22-27]

Mepolizumab is a fully humanized mouse anti-human IL-5 monoclonal antibody of the IgG1/ κ subtype, with a high molecular weight (149.2 kiloDaltons). The specific binding region of a humanized antibody is derived from a mouse anti-human IL-5 monoclonal antibody from which the complementary determining regions were grafted onto the heavy and light chain framework regions of a human IgG1/ κ antibody. Mepolizumab binds to IL-5 with high affinity and specificity.^[28] This prevents IL-5 from binding to the IL-5 receptor complex α -chain on the eosinophil cell surface, thereby blocking the actions of IL-5. Reslizumab is of the IgG4/ κ subtype and is based on a rat IgG2a antibody against human IL-5.^[24] The current review focuses on the pharmacokinetics and pharmacodynamics of mepolizumab from preclinical investigations and clinical studies in healthy subjects and patients with eosinophil-associated disorders.

2. In Vitro Studies

In *in vitro* experiments, mepolizumab inhibited binding of IL-5 to its receptor with a half-maximal inhibitory concentration (IC_{50}) of 0.94 nmol/L (range 0.70–1.2 nmol/L)^[29] and recognized human IL-5 with high affinity (dissociation

constant = 4.2 pmol/L).^[28,30] In addition, the stoichiometry of IL-5 and mepolizumab binding was shown to be 2:2, such that two IL-5 dimers are cross-linked by two molecules of mepolizumab.

Mepolizumab inhibited the proliferative responses of B13 cells (a murine IL-5/IL-3-dependent pre-B cell line) and TF-1.28 cells (a human erythroleukaemic cell line expressing functional IL-5 receptors) to recombinant human and monkey IL-5 with mean IC_{50} values of approximately 80 pmol/L.^[30] Mepolizumab also suppressed eosinophil differentiation induced by recombinant human or monkey IL-5 in both human (IC_{50} 70 pmol/L and 83 pmol/L, respectively) and monkey (IC_{50} 104 pmol/L and 116 pmol/L, respectively) bone marrow.^[30]

3. *In Vivo* Monkey Pharmacokinetic/Pharmacodynamic Studies

Mepolizumab antagonized the actions of monkey IL-5 as well as human IL-5 in cell culture; thus, cynomolgus monkeys were considered an appropriate *in vivo* model for assessing the pharmacology and long-term safety of mepolizumab.^[30]

Following intravenous injection, mepolizumab displayed dose-proportional pharmacokinetics over a dose range of 0.05–300 mg/kg.^[31] Plasma concentrations declined bi-exponentially, with a mean terminal half-life of 13.1 days. Mean plasma clearance and steady-state volume of distribution (V_{ss}) were 0.157 mL/h/kg and 65.6 mL/kg, respectively. After subcutaneous administration, mepolizumab was completely bioavailable, reaching maximal plasma concentrations 2–4 days post-injection.^[31] Concentrations decreased mono-exponentially after the initial absorption phase, with a mean terminal half-life of 14.5 days. The maximum pharmacodynamic response (81–96% decrease in peripheral eosinophil count from baseline levels) occurred 3 weeks after a single subcutaneous dose, and sustained suppression of peripheral blood eosinophils was observed following monthly subcutaneous dosing for 6 months. The subcutaneous pharmacokinetic/pharmacodynamic data fit an inhibitory indirect pharmacological response model with a mean IC_{50} for reduction in eosinophil count of 1.43 μ g/mL.

Additional studies in cynomolgus monkeys confirmed the pharmacodynamic effects of mepolizumab in terms of suppressing peripheral blood and bronchoalveolar lavage fluid (BALF) eosinophil counts in animals with IL-2-induced eosinophilia, and reducing blood and BALF eosinophil counts and levels of Regulated on Activation, Normal T Expressed and Secreted (RANTES) following antigen challenge.^[30]

The marked and sustained reductions in circulating and tissue eosinophils observed in these cynomolgus monkey stu-

dies suggested a potential therapeutic role for mepolizumab in diseases where eosinophil overproduction contributes to the pathophysiology.

Plasma concentrations of mepolizumab in both monkeys and humans were determined using a validated immunoassay. The lower limit of quantification for the assay was 50 ng/mL.

4. Pharmacokinetics in Healthy Subjects

Early assessments of the bioavailability, safety, tolerability and effect on eosinophil counts of mepolizumab were conducted in an open-label, single-dose, randomized, parallel-group, 12-week study (study SB-240563/018) involving 60 healthy subjects aged 20–52 years.^[32] Mepolizumab 250 mg was administered intravenously as a 30-minute infusion ($n=12$), as a subcutaneous injection in the abdomen, arm or thigh ($n=12$ for each site) or as an intramuscular injection in the lateral thigh ($n=12$). Subjects were followed up for 12 weeks, throughout which full-profile blood samples (i.e. at 2, 4, 6 and 8 hours post-dose, then daily for 1 week, and then at 2, 3, 4, 6, 8 and 12 weeks) were obtained for mepolizumab pharmacokinetic analyses and peripheral eosinophil counts.

Key pharmacokinetic parameters of mepolizumab in healthy subjects are summarized in table I. Compared with the intravenous route, the bioavailability of mepolizumab following subcutaneous injection in the abdomen, arm or thigh was 64%, 75% and 71%, respectively; bioavailability following intramuscular injection in the thigh was 81%. Plasma concentrations declined bi-exponentially after intravenous infusion and were representative of a two-compartment model (figure 1). Maximum plasma concentrations (C_{max}) occurred at 0.5–4.8 hours relative to the start of the infusion. Mepolizumab was well absorbed after subcutaneous and intramuscular injection, although time to reach C_{max} (t_{max}) was longer (range 2–14 days) compared with intravenous infusion. The terminal half-life of mepolizumab was approximately 20 days (regardless of the route of administration), which is consistent with the elimination half-life reported for endogenous IgG1 antibodies in humans.^[36,37]

In terms of the pharmacodynamic effect, peripheral blood eosinophil count data indicated a marked reduction of at least 50% from baseline.

5. Pharmacokinetics/Pharmacodynamics in Patients with Asthma

The pathophysiology of asthma is mediated, at least in part, by eosinophilic infiltration into pulmonary tissue and their subsequent degranulation and release of cytotoxic granule

Table 1. Summary of results from mepolizumab clinical pharmacokinetic studies with full sampling data

Study	Treatment	C _{max} (µg/mL) ^a	t _{max} (d) ^b	AUC _∞ (µg • d/mL) ^a	t _{1/2} (d) ^a	Plasma clearance (mL/h/kg) ^a	V _{ss} (mL/kg) ^a
Healthy subjects							
Study SB-240563/018: ^[32] open-label, randomized, parallel-group; 12 wk follow-up	250 mg, single dose						
	SC in abdomen (n = 12)	34.1 ± 12.1	7.0 [4.0–14.0]	1110 ± 372	17.9 ± 3.3		
	SC in arm (n = 12)	34.9 ± 7.3	5.0 [3.0–14.0]	1238 ± 228	20.4 ± 2.6		
	SC in thigh (n = 12)	38.2 ± 9.1	5.0 [2.0–7.0]	1196 ± 254	18.5 ± 3.5		
	IM (n = 12)	46.9 ± 10.6	4.0 [3.0–7.0]	1395 ± 348	19.2 ± 4.2		
	IV (n = 12)	109 ± 17	0.08 [0.02–0.2]	1557 ± 250	18.5 ± 2.3		
Single-dose studies in patients with asthma							
Study SB-240563/001: ^[33] multicentre, double-blind, randomized, placebo-controlled; 16 wk follow-up	Single IV dose						
	0.05 mg/kg (n = 4)	1.03 ± 0.21	0.04 [0.03–0.08]	15.5 ± 2.7	19.7 ± 7.6	0.137 ± 0.022	85.0 ± 19.8
	0.5 mg/kg (n = 5)	10.6 ± 2.2	0.08 [0.03–0.13]	168 ± 19	20.0 ± 1.9	0.125 ± 0.014	82.2 ± 14.5
	2.5 mg/kg (n = 8)	51.4 ± 7.9	0.04 [0.02–0.08]	846 ± 164	19.3 ± 2.9	0.129 ± 0.023	80.6 ± 7.8
	10 mg/kg (n = 8)	215 ± 28	0.04 [0.02–0.08]	3345 ± 324	19.0 ± 2.5	0.126 ± 0.013	78.9 ± 11.0
Study SB-240563/035: ^[34] double-blind, randomized, placebo-controlled, dose-rising; 16 wk follow-up	Single IV dose						
	0.5 mg/kg (n = 4)	12.1 ± 2.4	0.03 [0.02–0.08]	207 ± 34	20.9 ± 4.0	0.103 ± 0.017	68.4 ± 2.5
	2.5 mg/kg (n = 4)	79.0 ± 4.3	0.06 [0.02–0.17]	1327 ± 247	21.7 ± 2.8	0.081 ± 0.015	55.4 ± 5.2
	10 mg/kg (n = 4)	278 ± 29	0.08 [0.02–0.34]	4361 ± 168	20.9 ± 2.6	0.096 ± 0.004	59.3 ± 3.7
Multiple-dose study in patients with asthma							
Study SB-240563/017: ^[35] double-blind, randomized, placebo-controlled, parallel-group; 16 wk follow-up	3 SC doses of 250 mg (day 0, wk 6 and wk 8)						
	Dose 1 (n = 8)	17.7 ± 7.1	4.5 [3.0–7.0]	560 ± 197	20.5 ± 5.3		
	Dose 3 (n = 8)	32.2 ± 7.8	8.0 [2.0–14.0]	924 ± 189	16.2 ± 2.1		

a Data are expressed as mean ± SD.

b Data are expressed as median [range].

AUC_∞ = area under plasma concentration-time curve from time zero to infinity; C_{max} = maximum plasma concentration; IM = intramuscular; IV = intravenous; SC = subcutaneous; t_{1/2} = half-life; t_{max} = time to reach C_{max}; V_{ss} = steady-state volume of distribution.

proteins.^[9,38,39] A correlation has been noted between airway eosinophil counts and features of asthma severity.^[40] In addition, when a management strategy aimed at normalizing the induced sputum eosinophil count is followed, significant reductions in asthma exacerbations, hospital admissions and rescue courses of oral corticosteroids are observed compared with patients managed in accordance with standard asthma guidelines.^[41] Clinical studies have also shown that IL-5 levels are increased in bronchial biopsies from patients with asthma,^[42] concentrations of IL-5 in bronchial tissue correlate with clinical features of asthma,^[43] and IL-5 mRNA is upregulated in bronchial mucosa after allergen challenge.^[44] Inhibition of eosinophil development or function through IL-5 blockade might thus be expected to have a beneficial therapeutic effect in asthma by preventing eosinophil maturation, function or migration into pulmonary tissue.

5.1 Single-Dose Studies

Mepolizumab was first evaluated in male patients, 18–45 years of age, with mild atopic asthma (forced expiratory volume in 1 second [FEV₁] ≥ 70% predicted and on β₂-adrenergic receptor agonists) in a multicentre, double-blind, randomized, placebo-controlled, dose-rising study (study SB-240563/001).^[45] Participants received a single intravenous infusion of mepolizumab 0.05 mg/kg (n = 4), 0.5 mg/kg (n = 5), 2.5 mg/kg (n = 8) or 10 mg/kg (n = 8), or placebo (n = 13). Mepolizumab was quantifiable in full-profile blood samples obtained on the day of the infusion and over the 16-week follow-up period in the majority of patients receiving doses >0.05 mg/kg. Mepolizumab plasma concentrations declined bi-exponentially after infusion (figure 2). A mean initial half-life of approximately 2 days was followed by a long terminal half-life of approximately 20 days (range

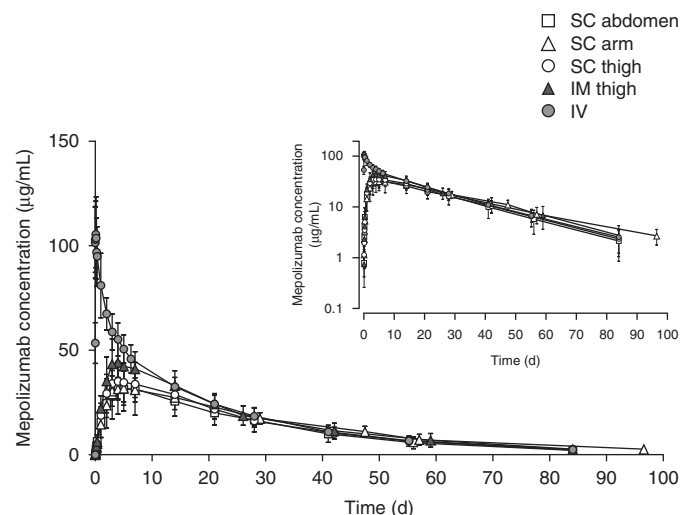


Fig. 1. Mean ($\pm$ SD) mepolizumab plasma concentration-time profiles following a single 250 mg dose administered as an intravenous (IV) infusion ($n=12$), as a subcutaneous (SC) injection in the abdomen, arm or thigh ($n=12$ for each site) or as an intramuscular (IM) injection in the lateral thigh ($n=12$) [study SB-240563/018].^[32] The inset shows the same data as a semilogarithmic plot.

14–30 days), which was independent of dose. A dose-proportional relationship was evident in the C_{max} and in the area under the plasma concentration-time curve from time zero to infinity (AUC_{∞}) [table I]; intersubject variability in these parameters was low, with coefficients of variation generally $<20\%$. Plasma clearance and V_{ss} were similar across the dose range, indicating linear pharmacokinetics.

Pharmacodynamic parameters in this study were assessed for mepolizumab doses of 2.5 and 10 mg/kg. Mean blood eosinophil counts decreased from baseline by 73% on day 8 and by 87% on day 29 with mepolizumab 10 mg/kg compared with 32% and 34%, respectively, with placebo.^[45] Mepolizumab 10 mg/kg also significantly attenuated the increase in blood eosinophils observed post-allergen challenge ($p < 0.001$) and reduced post-allergen challenge sputum eosinophils ($p < 0.01$) compared with placebo. Less profound effects on blood and sputum eosinophil numbers were observed with mepolizumab 2.5 mg/kg.

A second double-blind, randomized, placebo-controlled, single-dose study (study SB-240563/035) was conducted in men, 18–43 years of age, with mild allergic asthma ($FEV_1 > 70\%$ predicted and on β_2 -adrenergic receptor agonists with or without inhaled corticosteroids at 400–1000 $\mu\text{g/day}$).^[34] Participants received a single intravenous infusion of mepolizumab 0.5 mg/kg ($n=4$), 2.5 mg/kg ($n=4$) or 10 mg/kg ($n=4$), or placebo ($n=6$). Mepolizumab plasma concentrations were approximately dose proportional and decreased bi-exponentially according to a two-compartment model. All pharmacokinetic parameters

were consistent with those reported in study SB-240563/001 (table I), and intersubject variability was low (coefficients of variation $\leq 20\%$). A persistent and dose-dependent reduction in peripheral blood eosinophil count relative to baseline was observed in the majority of patients receiving mepolizumab, but not in those receiving placebo (figure 3). The duration of the suppression in cell count varied among dose groups and appeared to be longer in the majority of patients who received either 2.5 or 10 mg/kg. The eosinophil count in all patients receiving 10 mg/kg remained suppressed until the follow-up visit at 16 weeks post-dose (8–36% of baseline levels), whereas the count was closer to baseline or, in two cases, higher than baseline at follow-up in patients receiving 0.5 mg/kg. The relationship between mepolizumab plasma concentrations and percentage changes in eosinophil count relative to baseline was described using an indirect pharmacological response model. A persistent and dose-dependent reduction in blood eosinophil count relative to baseline was observed in most mepolizumab-treated patients (estimated maximal decrease of approximately 85%; IC_{50} of 0.45 $\mu\text{g/mL}$). The duration of eosinophil suppression increased with increasing dose, with the maximal reduction in eosinophil count occurring 3–4 days after C_{max} .

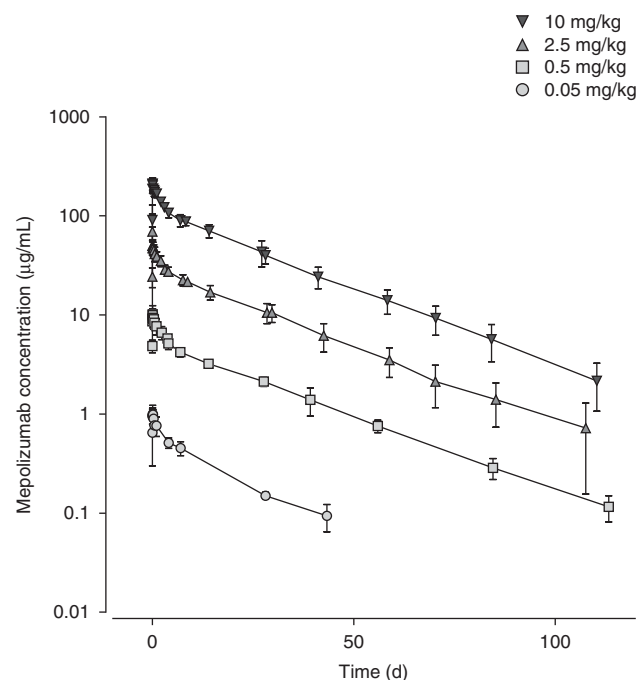


Fig. 2. Semilogarithmic plot of mepolizumab plasma concentrations (mean $\pm$ SD) vs time following a single intravenous infusion of mepolizumab 0.05 mg/kg ($n=4$), 0.5 mg/kg ($n=5$), 2.5 mg/kg ($n=8$) or 10 mg/kg ($n=8$), or placebo ($n=13$) to male patients with mild atopic asthma (study SB-240563/001).^[33,45]

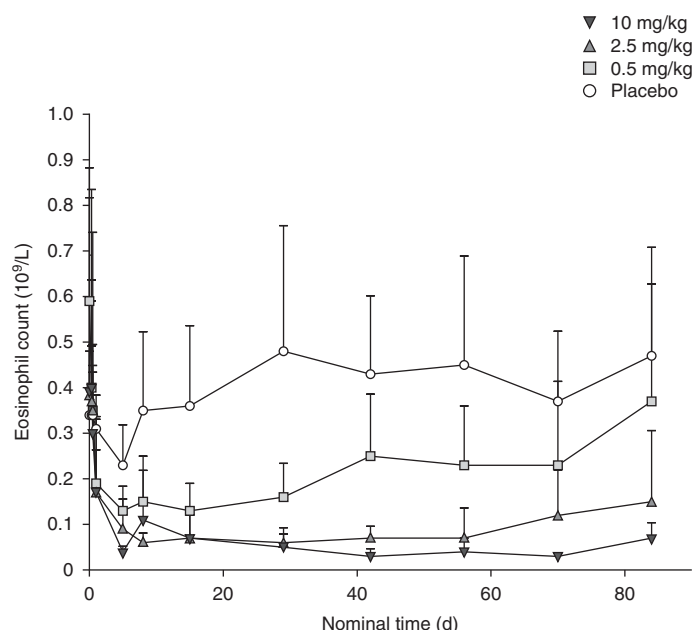


Fig. 3. Peripheral blood eosinophil count (mean + SD) vs time profiles in patients with mild atopic asthma following single intravenous doses of 0.5, 2.5 or 10 mg/kg mepolizumab or placebo (n = 4 per treatment group) [study SB-240563/035].^[34]

5.2 Multiple-Dose Studies

Men and women, 19–50 years of age, with mild asthma ($FEV_1 > 70\%$ predicted and on β_2 -adrenergic receptor agonists with or without inhaled corticosteroids at $\leq 1000 \mu\text{g/day}$ [or fluticasone propionate at $\leq 500 \mu\text{g/day}$]) received three abdominal subcutaneous doses of mepolizumab 250 mg (n = 8) or placebo (n = 8) in a double-blind, randomized, placebo-controlled, parallel-group study (study SB-240563/017).^[35] Doses 1 and 2 were given 6 weeks apart; dose 3 was given 2 weeks after dose 2. Mepolizumab was quantifiable in all blood samples collected throughout the entire study period (to 16 weeks after the third dose; week 24). As expected, based on the short time intervals between the 3 doses (6 and 2 weeks, respectively) relative to the long terminal half-life of mepolizumab (~ 20 days), plasma accumulation was observed following the third dose. AUC_{∞} was 65% higher after dose 3 than after dose 1 and C_{max} was 80% higher after dose 3 than after dose 1 (table I). Values of t_{max} and terminal half-life were consistent with those following the administration of single subcutaneous doses of mepolizumab 250 mg to healthy subjects (study SB-240563/018) [table I]. Pharmacokinetic modelling suggested that mepolizumab exhibited time-independent pharmacokinetics. With regard to pharmacodynamics, a reduction in peripheral blood eosinophils of $\sim 70\%$ relative to baseline was observed after the third

dose of mepolizumab and counts remained suppressed for most of the 16-week follow-up period. No reduction in blood eosinophils was observed among patients receiving placebo.

The pharmacokinetics of mepolizumab following multiple intravenous infusions have also been assessed. This multicentre, double-blind, randomized, placebo-controlled, parallel-group study included men and women, 18–55 years of age, with persistent mild-to-moderate asthma (FEV_1 50–80% predicted, managed with inhaled beclomethasone $\leq 1000 \mu\text{g/day}$ or equivalent) [study SB-240563/006].^[46] Patients received three intravenous infusions of mepolizumab 250 mg (n = 120) or 750 mg (n = 116), or placebo (n = 126) at intervals of 1 month. Sparse blood samples for pharmacokinetic analyses from designated subgroups of subjects at different time points were obtained over the 12-week treatment period and 8-week follow-up period. Mepolizumab was quantifiable in plasma at all study visits after the first infusion, consistent with its long terminal half-life; concentrations at each visit were approximately dose-proportional between the 250 mg and 750 mg doses. Mean ($\pm$ SD) plasma concentrations 4 weeks after the third dose were $22.2 \pm 9.1 \mu\text{g/mL}$ in the 250 mg dose group and $51.2 \pm 19.5 \mu\text{g/mL}$ in the 750 mg dose group. Mepolizumab plasma concentrations at each visit were similar to predicted multiple-dose concentration data simulated from a single-dose pharmacokinetic study (study SB-240563/001) in patients with mild asthma, suggesting time- and concentration-independent pharmacokinetics.

Both mepolizumab doses reduced mean blood eosinophils by $\sim 80\%$ ($p < 0.001$ vs placebo). Decreases were apparent 1 week after the first infusion and decreased further by week 4. The reduction was maintained for 12 weeks after the last infusion (to week 20) with mepolizumab 750 mg, although eosinophil counts began to rise after week 16 with mepolizumab 250 mg (figure 4). Eosinophil numbers in induced sputum also decreased (n = 37) by 1 week after the first infusion and persisted to the final follow-up visit at week 20; the reduction from baseline to week 12 was statistically significant for mepolizumab 750 mg ($p = 0.013$), with a trend towards significance for mepolizumab 250 mg ($p = 0.061$). No significant changes in blood or sputum eosinophil levels were detected with placebo.

The effects of multiple intravenous doses of mepolizumab on eosinophil levels in various tissues in patients with mild atopic asthma ($FEV_1 \geq 70\%$ predicted, on as-needed inhaled short-acting β_2 -adrenergic receptor agonists only) have also been evaluated (study SB-240563/036).^[38,47] Three doses of mepolizumab 750 mg (n = 11) or placebo (n = 13) were given 4 weeks apart in this double-blind, randomized, parallel-group study. Blood eosinophil counts were significantly lower in the mepo-

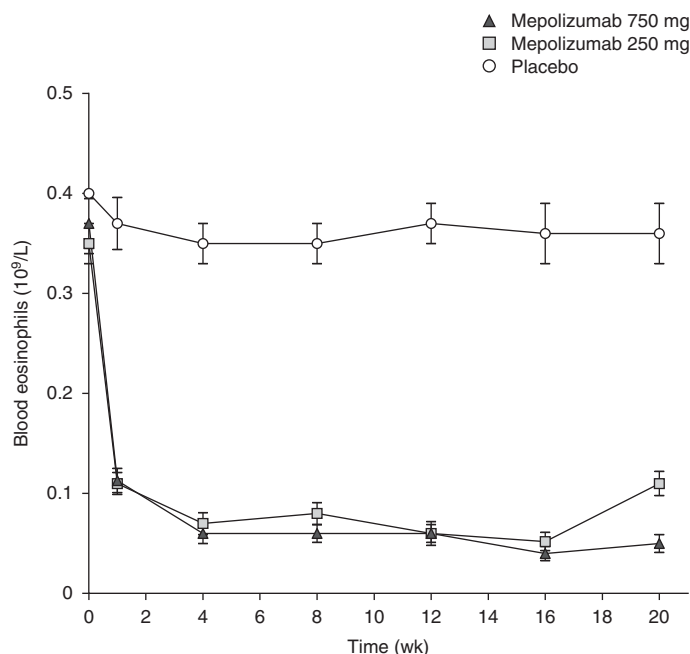


Fig. 4. Peripheral blood eosinophil count (mean $\pm$ standard error of mean [SEM]) vs time in patients with mild asthma following intravenous infusion of three doses of mepolizumab 250 mg ($n=120$) or 750 mg ($n=116$), or placebo ($n=126$) at intervals of 4 weeks (study SB-240563/006). [Reprinted from Flood-Page et al.,^[46] with permission of the American Thoracic Society. Copyright © American Thoracic Society.]

lizumab group than in the placebo group at week 4 ($p=0.002$) and week 9 ($p=0.02$), with a median reduction of 100% from baseline (interquartile range [IQR] 67–100%; figure 5).^[38] Significant reductions were also seen in the number of eosinophils within bronchial mucosal biopsies (median reduction of 55% from baseline to week 9 [IQR 29–89%]; $p=0.009$ vs placebo) and bone marrow eosinophils (median reduction of 52% from baseline to week 9 [IQR 45–76%]; $p=0.003$ vs placebo) [figure 5].^[38] Eosinophil counts in nasal lavage samples decreased significantly from baseline to week 9 both pre- ($p=0.006$ vs placebo) and 6 hours post- ($p=0.027$ vs placebo) nasal allergen challenge.^[48] Similarly, cutaneous eosinophil counts decreased significantly with mepolizumab versus placebo at 6 hours ($p=0.0015$) and 48 hours ($p=0.0025$) following intradermal allergen challenge.^[47] No statistically significant reductions were observed in BALF eosinophil levels from baseline to week 9 (median reduction of 79% from baseline [IQR 43–99%]; $p=0.4$ vs placebo).^[38] Suppression of eosinophil numbers in blood and sputum have also been reported in patients with refractory eosinophilic asthma (defined as sputum eosinophilia $>3\%$ despite systemic corticosteroid treatment) and a history of severe exacerbations^[49] and in those with corticosteroid-dependent asthma with sputum eosinophilia.^[50]

6. Pharmacokinetics/Pharmacodynamics in Patients with Hypereosinophilic Syndrome

HES is characterized by sustained eosinophilia (blood eosinophil count >1500 cells/ μL for >6 months), without a

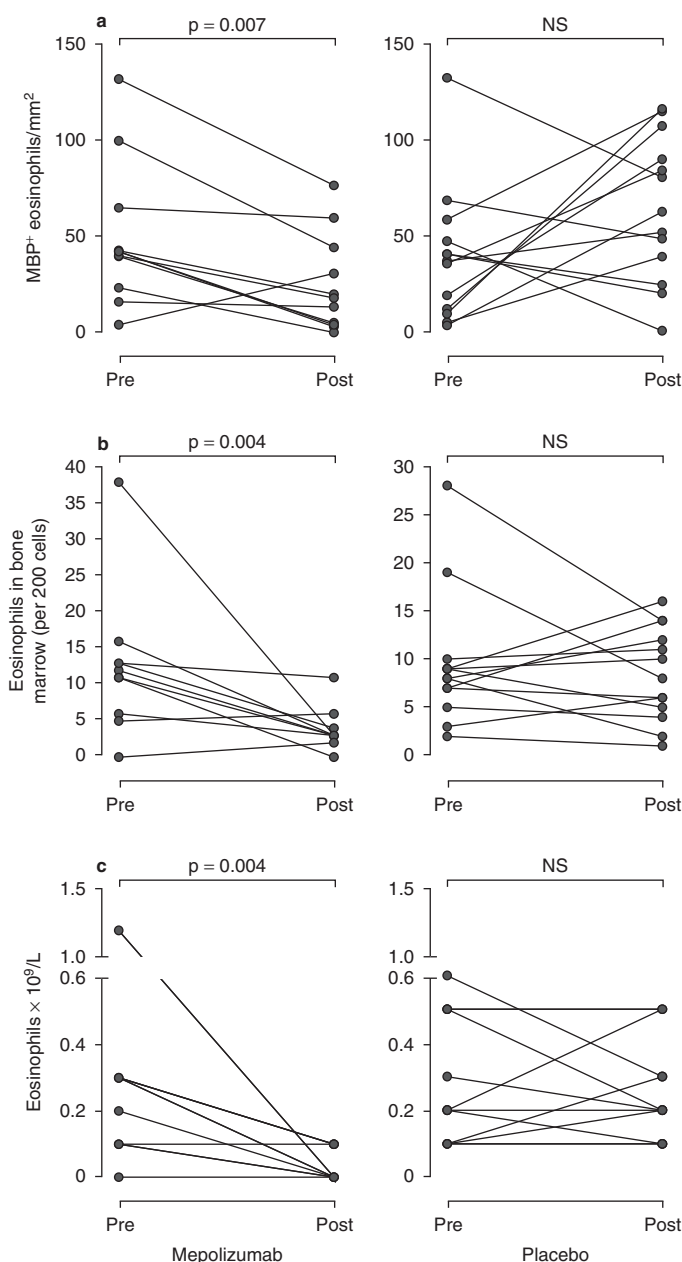


Fig. 5. Effect of mepolizumab or placebo on eosinophils within the (a) bronchial mucosa, (b) bone marrow and (c) blood in patients with mild asthma after three intravenous doses of mepolizumab 750 mg ($n=11$) or placebo ($n=13$) at intervals of 4 weeks. The post-treatment time point was 1 week after the third dose (study SB-240563/036). MBP+ = major basic protein-positive; NS = not significant. [Reprinted from Flood-Page et al.,^[38] with permission of the American Thoracic Society. Copyright © American Thoracic Society.]

secondary cause and with evidence of end-organ involvement with eosinophil infiltration and injury.^[51] Existing therapies have significant limitations – including serious adverse effects, poor long-term tolerability (impacting quality of life), the development of resistance and a need for regular monitoring – and they are not universally effective in all patients.^[11] Since IL-5 appears to contribute to the pathogenesis of some HES phenotypes,^[52] IL-5 inhibition is a logical therapeutic target for this disease.

A multicentre, international, randomized, double-blind, placebo-controlled trial (study MHE100185) has been conducted in 85 patients with HES (negative for the *FIPILI/PDGFR* mutation) maintained on prednisone 20–60 mg/day.^[53] Participants received intravenous mepolizumab 750 mg (n=43) or placebo (n=42) at 4-week intervals for 32 weeks (8 doses) while tapering the prednisone dose according to a predefined algorithm based on blood eosinophil counts and clinical manifestations of HES. Mepolizumab was quantifiable in all plasma samples collected after the first infusion (day 1 post-infusion and at weeks 1, 2, 3, 4, 8, 16, 24, 32 and 36).

Mean mepolizumab plasma concentrations were similar to predicted multiple-dose concentrations estimated from a two-compartment intravenous infusion pharmacokinetic model based on a previous single-dose study (study SB-240563/035^[34]) in male patients with mild asthma (figure 6).

The pharmacodynamic effect of mepolizumab in patients with HES was manifested as suppression of the eosinophil count to <600 cells/ μ L for ≥ 8 consecutive weeks. This endpoint was achieved in 95% of mepolizumab-treated patients versus 45% of placebo-treated patients (odds ratio 18.9 [95% CI 4.7, 75.2]; $p < 0.001$). Case reports and an open-label study have also documented sustained reductions in blood eosinophils in patients with HES.^[23,55]

7. Pharmacokinetics/Pharmacodynamics in Patients with Eosinophilic Oesophagitis

Eosinophilic oesophagitis is characterized by symptoms such as food impaction and dysphagia and dense oesophageal eosinophilia (≥ 15 eosinophils/high-power field [hpf] in biopsy specimens) after exclusion of other disorders associated with similar clinical, histological or endoscopic features.^[56] Experimental murine models of eosinophilic oesophagitis suggest a major role for IL-5 in the pathogenesis of this disease. For example, eosinophilic oesophagitis can be induced by IL-5 overexpression in transgenic mice^[57] and blocked by neutralizing IL-5 in an allergen-induced murine model of eosinophilic oesophagitis,^[58] and IL-5-deficient mice are protected from

induction of experimental eosinophilic oesophagitis.^[58,59] Thus, a monoclonal antibody that binds to and inactivates IL-5 might be expected to have efficacy in relieving the symptoms of eosinophilic oesophagitis.

The pharmacokinetics of mepolizumab were assessed in adults with histologically active (>20 eosinophils/hpf), severe eosinophilic oesophagitis unresponsive to or dependent on corticosteroids in a randomized, placebo-controlled, double-blind pilot trial (study MEE103226).^[60] Two intravenous doses of mepolizumab 750 mg (n=5) or placebo (n=6) were administered 1 week apart; as no patient was in complete remission (<5 eosinophils/hpf) at week 4, all received 2 additional doses, at weeks 5 and 9, of mepolizumab 1500 mg or placebo. Mepolizumab was quantifiable in plasma at all visits after the first infusion. Concentrations following infusion of mepolizumab 750 mg or 1500 mg were similar to values predicted from the antibody's pharmacokinetic behaviour after single doses in male patients with mild asthma (study SB-240563/035^[34]). The good predictability of multiple-dose concentrations at 1500 mg doses suggests that mepolizumab has time- and concentration-independent pharmacokinetics at doses twice the proposed clinical dose of 750 mg.

Mean reductions from baseline in blood eosinophil counts were 82% at week 4 and 89% at week 13 with mepolizumab compared with 17% and 24%, respectively, with placebo. Histological assessments showed reductions from baseline in mean and peak oesophageal eosinophil levels with mepolizumab at

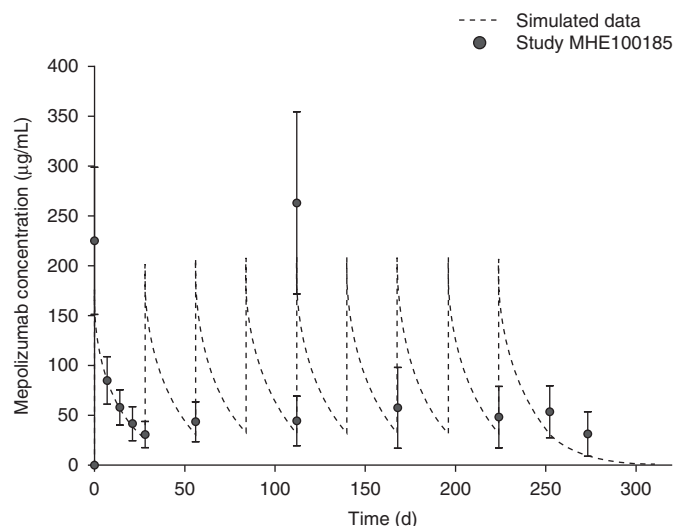


Fig. 6. Observed mepolizumab plasma concentrations (mean $\pm$ SD) in patients with hypereosinophilic syndrome (study MHE100185)^[53,54] after intravenous infusion of mepolizumab 750 mg at intervals of 4 weeks (n=43) overlaid with predicted concentrations simulated from a single-dose study (study SB-240563/035)^[34] in patients with mild asthma.

week 4 (59% and 62%, respectively) and at week 13 (71% and 59%, respectively). In the placebo group, mean oesophageal eosinophils decreased from baseline by 13% at week 4 and 1% at week 13; peak eosinophils decreased by 23% at week 4 but increased by 12% by week 13.

Reductions in blood and oesophageal eosinophils have also been reported in two open-label studies in patients with eosinophilic oesophagitis.^[23,61]

8. Pharmacodynamics in Patients with Atopic Dermatitis

Blood and tissue eosinophil numbers and granule products are increased in most patients with atopic dermatitis and correlate approximately with disease severity.^[2] The dermal inflammatory infiltrate in atopic dermatitis lesions contains eosinophils and cytotoxic eosinophilic granule proteins^[2,62] and, in experimental models, eosinophils are present at lesion sites before symptoms appear.^[2] Interventions that lead to clinical improvement are associated with decreased eosinophilic inflammation. Because of the pivotal role of IL-5 in eosinophilia and its selectivity for eosinophils, neutralization of IL-5 may offer benefit in patients with atopic dermatitis.

Mepolizumab reduced blood eosinophils in adult patients with moderate-to-severe atopic dermatitis in a double-blind, randomized, placebo-controlled, parallel-group study (study SB-240563/045).^[22] Participants received two infusions of mepolizumab 750 mg ($n = 20$) or placebo ($n = 23$), with 1 week between doses. A significantly greater mean reduction from baseline in blood eosinophil counts occurred with mepolizumab compared with placebo at week 2 (difference of -489 cells/ μL [90% CI $-648, -329$]; $p < 0.001$). However, there was no effect on dermal eosinophil levels, possibly because residual circulating eosinophils may continue to migrate into cutaneous tissue in response to other mediators, such as eotaxin and RANTES. The pharmacokinetics of mepolizumab were not evaluated in this study.

9. Mepolizumab Elimination

The exact route or routes of elimination are not known. As mepolizumab is a monoclonal antibody, no mass balance study was conducted. The pharmacokinetics (including the half-life) of mepolizumab, a fully humanized monoclonal antibody, are very similar to those of IgG1 antibodies; therefore, it is probably eliminated in a manner similar to IgG1. The vast majority of immunoglobulins are catabolized by proteases with protection by the endothelial receptor FcRn.^[63] The Fc fragment of mepolizumab is fully humanized and should have affinity for

the FcRn receptor similar to that of IgG1 with an equally long half-life, which is consistent with the data presented in the previous sections.^[64]

Data in the monkey add credence to the proposed elimination pathway. The half-life in monkeys is shorter than in humans, being approximately 2 weeks versus 3 weeks. The affinity of the humanized Fc fragment for the monkey FcRn receptor is weaker than in humans, resulting in shorter half-life values in monkeys versus humans.^[65]

10. Patients with Hepatic or Renal Impairment

Across the programme, there have been no patients with hepatic impairment for whom mepolizumab concentration data were available. As mepolizumab is probably eliminated by ubiquitous proteolytic enzymes, there should be no difference in the pharmacokinetics of mepolizumab in patients with hepatic impairment versus those without hepatic impairment.

Sixteen HES patients had creatinine clearance values < 80 mL/min and received mepolizumab, with five of those subjects having creatinine clearance values < 60 mL/min. Mepolizumab plasma concentrations were similar among these patients and those with normal renal function. This finding is not unexpected for an immunoglobulin that has a high molecular weight, such as mepolizumab, as compounds with high molecular weights are not filtered by the kidney.

There are no data in patients who have renal or other diseases that cause proteinuria such that IgG is eliminated in the urine.

11. Effect of Age or Sex

Eight patients with HES receiving mepolizumab were over 65 years of age (range 66–74 years). Mepolizumab concentrations in these patients were similar to the mean for the overall population in this study.

Approximately 40% ($n = 17$) of the patients in the HES trial were females. In general, plasma mepolizumab concentrations in these subjects were similar to those in males.

Both of these findings are similar to those observed with other monoclonal antibodies.

12. Safety with Respect to Dose or Pharmacodynamics

In general, mepolizumab was well tolerated at all doses studied. In the single-dose, dose-escalation studies, there were no deaths or serious adverse events reported. The incidence of

adverse events was comparable between patients treated with placebo or mepolizumab at different dose levels or routes.

In multiple-dose, placebo-controlled studies, the safety profile was similar between the placebo group and the mepolizumab group. In study SB-240563/006,^[46] two doses of mepolizumab (250 mg and 750 mg) and placebo were administered in a parallel design. The overall incidence of adverse events was greater in the placebo group than in the mepolizumab treatment groups, with no difference between the two treatment dose groups. In HES study MHE100185,^[53] the overall incidence of adverse events was similar between the mepolizumab and placebo groups (93% and 98%, respectively). Serious adverse events were also reported in a similar number of patients. There was one death in a mepolizumab-treated patient who had a history of severe HES and multiple co-existing cardiovascular conditions; the event was considered unrelated to treatment with mepolizumab by the investigator.

Mepolizumab was generally well tolerated in all trials in all diseases studied. No serious risks associated with mepolizumab or clear predisposing factors for adverse events were identified.

In all treatment groups, considerable decreases in eosinophils were observed, as expected. Depending on the disease, eosinophils were either in the normal range or elevated (HES) in the placebo groups.

13. Development of Mepolizumab

In the early asthma studies, doses of mepolizumab were administered on a mg/kg basis. Review of the pharmacokinetic data from these studies indicated the variability in the pharmacokinetic parameters was similar if data were analysed for a fixed dose or body size-based dosing. Due to ease of dose preparation, reduced cost, and reduced chance of dosing errors, further trials in adults were, and are, being conducted with a fixed-dose regimen. Evaluation of monthly intravenous and subcutaneous doses ranging from 75 mg to 750 mg is ongoing.

The final dose and dosing interval will be disease- and possibly patient-specific. For example, in the HES population, all patients receiving active treatment (43 of 85 in total) in the pivotal trial received nine monthly 750 mg intravenous doses.^[53] Following the original study, patients could enter an extension study (study MHE100901).^[66] At the time of the interim analysis, 61 of the original 78 patients enrolled in the extension study had a mean exposure to mepolizumab of 20.6 months (range 1–35 months). The majority of patients received infusions at intervals of 5–16 weeks during the phase when the dose frequency was individualized.

In addition to the disease states previously discussed, mepolizumab has been evaluated in paediatric eosinophilic oesophagitis and in collaborative research trials (i.e. investigator-initiated studies) in Churg-Strauss syndrome, eosinophilic bronchitis, severe uncontrolled refractory asthma, nasal polyposis and HES.

Two diseases for which mepolizumab is being evaluated, HES and Churg-Strauss syndrome, are rare diseases that can be severe and life threatening and are often associated with significant morbidity. This is, in part, due to the present standard of care, corticosteroids and immunomodulators. Clinical data in both HES and Churg-Strauss syndrome have shown that mepolizumab allows for decreased doses of corticosteroids or no corticosteroids while maintaining clinical stability.^[53,67] Given the substantial side-effect burden for patients undergoing long-term corticosteroid therapy, including increased risk of infection, adrenal suppression, weight gain, thinning skin, osteoporosis, cataracts, glaucoma, hyperglycaemia and depression, mepolizumab, which has few to no side effects, would be of significant clinical benefit.

14. Conclusions

The pharmacokinetics of the anti-IL-5 monoclonal antibody, mepolizumab, have been evaluated in animal models (cynomolgus monkeys) and human subjects across a wide range of doses and by different routes of administration (intravenous, subcutaneous and intramuscular). Mepolizumab exhibited dose-proportional and time-independent pharmacokinetics consistently in all *in vivo* and clinical studies, at all doses and in all populations examined. Plasma concentrations declined in a bi-exponential manner following intravenous administration of single doses to monkeys, healthy subjects or patients with asthma, and were well described using a two-compartment model. Estimates based on single-dose clinical studies predicted mepolizumab plasma concentrations in multiple-dose studies involving patients with asthma, HES or eosinophilic oesophagitis.

The pharmacokinetic profile of mepolizumab, which is comparable to those of other large-molecular-weight monoclonal antibodies,^[37,63,68] was generally similar in monkeys and humans. Following intravenous infusion of doses >0.05 mg/kg, mepolizumab was quantifiable in human plasma for up to 16 weeks post-dose. The mean initial half-life of approximately 2 days was followed by a long mean terminal half-life of approximately 20 days; these half-life values for humans were slightly longer than those observed in monkey models (12.9 hours and 13.1 days, respectively). Plasma clearance

values (0.064–0.163 mL/h/kg) indicated that elimination was slow, and the V_{ss} (49–93 mL/kg) represented about 1.5–2 times plasma volume. Bioavailability in humans was 64–75% following subcutaneous injection (dependent on the administration site) and 81% following intramuscular injection.

Pharmacodynamically, mepolizumab produced persistent reductions in blood eosinophil counts relative to baseline after intravenous, subcutaneous or intramuscular administration to healthy subjects or patients across all severities of asthma, HES, eosinophilic oesophagitis, atopic dermatitis or atopic asthma. The maximal reduction in eosinophil count occurred 3–4 days after infusion and was estimated at ~85% relative to baseline using population modelling; the IC_{50} was 0.45 µg/mL. The duration of eosinophil suppression was dependent on the mepolizumab plasma concentration. An indirect pharmacological response model best described the relationship between the reduction in peripheral eosinophil counts and mepolizumab plasma concentrations. Decreases in eosinophil counts were also observed in the oesophagus, induced sputum, skin (post-allergen challenge), bone marrow, nasal lavage fluid (post-allergen challenge) and bronchial mucosa biopsy in mepolizumab-treated patients compared with placebo groups.

Given its mechanism of action, mepolizumab has the potential to provide a new therapeutic option for subjects with eosinophilic disorders. In particular, early trials in severe asthma patients support a role for anti-IL-5 treatment; additional clinical trials will further assess the potential for mepolizumab in the treatment of such patients.

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